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Invention Disclosure Format (IDF)-B

1. Title of the invention:

CryptoGuard AI: A Multi-Model AI-Driven System and Method for Cryptographic Vulnerability Detection, NIST SP 800-90B Compliant Entropy Analysis with Leftover Hash Lemma Extraction, and Post-Quantum Migration Guidance Using Large Language Models on Cloud GPU Infrastructure

2. Field / Area of invention:

The present invention pertains to the interdisciplinary fields of **Artificial Intelligence (AI)**, **Cryptographic Security**, and **Software Engineering**. Specifically, it relates to:

- The application of large language models (LLMs) for automated static analysis of cryptographic code vulnerabilities with deep semantic reasoning;
- Statistical entropy quality assessment compliant with NIST SP 800-90B standards, integrated with AI-powered natural language interpretation;
- Provably secure randomness extraction via the Leftover Hash Lemma with HKDF-SHA3-256 conditioning;
- Intelligent post-quantum cryptography migration advisory systems targeting NIST-standardized algorithms (FIPS 203, 204, 206).

The invention falls within the domain of AI-assisted cybersecurity tools, entropy source validation, and computer-implemented methods for cryptographic assurance.

International Patent Classification (IPC):

- **G06F 21/57** — Certifying or testing software for faults (vulnerability detection)
- **G06N 3/08** — Machine learning methods — Large Language Models (multi-model routing)
- **H04L 9/08** — Key distribution including post-quantum cryptography (PQ migration)
- **G06F 7/58** — Random number generation and testing (entropy analysis)
- **H04L 9/32** — Authentication / integrity (hash function analysis, HMAC verification)

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

S.No.	Prior Art / Reference	Year	Type	Key Contribution	Limitation / Gap Addressed by Present Invention
1	Snyk Code (Snyk Ltd., London, UK)	2019	Commercial Tool	AI-powered static analysis using rule-based pattern matching and ML classifiers for security vulnerability detection.	Limited to predefined rule sets; no deep semantic reasoning about cryptographic correctness. Cannot assess entropy quality, apply LHL, or provide post-quantum migration guidance. Unable to explain *why* a finding is cryptographically dangerous.
2	SonarQube (SonarSource SA, Geneva, Switzerland)	2007	Commercial Tool	Continuous code quality inspection with security hotspot detection for 30+ programming languages.	Pattern-matching only; lacks AI-driven reasoning. No cryptography-specific taxonomy. No entropy analysis module, no LHL extraction, no PQ algorithm recommendations.
3	NIST SP 800-90B, "Recommendation for the Entropy Sources Used for Random Bit Generation" (Turan, M.S. et al.)	2018	Standard / Publication	Federal standard for entropy source validation with min-entropy estimation methods including MCV, collision, Markov, and compression estimators.	Provides statistical methods only; no AI interpretation of results. No integration with code scanning tools. Requires expert knowledge to interpret numerical outputs. No automated remediation or key derivation.
4	Dieharder v3.31 (Brown, R.G.) / ENT (Walker, J.)	2004 / 1998	Open Source Tool	Statistical test suites for RNG quality assessment including chi-squared, serial correlation, birthday spacings, and spectral tests.	Command-line tools producing raw numerical output. No AI interpretation, no visual heatmaps, no code vulnerability scanning integration. Requires statistical expertise for meaningful interpretation.

S.No.	Prior Art / Reference	Year	Type	Key Contribution	Limitation / Gap Addressed by Present Invention
5	GitHub Copilot (Microsoft Corp. / OpenAI Inc.)	2021	Commercial Tool	LLM-powered code completion and conversational assistant using GPT-4 for general programming tasks across all domains.	General-purpose coding assistant with no cryptography-specific vulnerability taxonomy. No NIST-compliant entropy analysis, no LHL integration, no structured PQ migration paths. Cannot produce severity-ranked security findings.
6	US Patent 11,379,588 B2 (IBM Corp.) — "Identifying Cryptographic Algorithms"	2022	Patent	System for detecting cryptographic algorithm usage in codebases and flagging deprecated or non-compliant algorithms via static analysis rules.	Rule-based detection only; no LLM-driven semantic analysis. No entropy quality assessment. No Leftover Hash Lemma integration. No multi-model routing or latency optimization. Single-purpose detection without remediation.
7	US Patent 10,846,410 B2 (Qualcomm Inc.) — "Entropy Source Validation"	2020	Patent	Hardware-level entropy source testing and validation for cryptographic random number generators in SoC designs.	Hardware-specific implementation; no software code analysis capability. No AI interpretation of entropy metrics. No integration with vulnerability scanning, PQ guidance, or LHL-based extraction.
8	Leftover Hash Lemma (Impagliazzo, R., Levin, L.A., Luby, M.), *STOC '89*	1989	Academic Publication	Foundational theorem for provably secure randomness extraction: if $H_{inf}(X) \geq k + 2*\log(1/\varepsilon)$, a universal hash extracts k near-uniform bits within ε statistical distance.	Purely theoretical result; no practical software implementation integrated with AI-driven entropy assessment. No automated k_eff computation pipeline or HKDF conditioning stage for deployable key material.
9	NIST FIPS 203/204/206 — ML-KEM, ML-DSA, SLH-DSA (NIST PQC Standardization)	2024	Standard	First NIST-standardized post-quantum cryptographic algorithms: CRYSTALS-Kyber (FIPS 203), CRYSTALS-Dilithium (FIPS 204), FALCON/SLH-DSA (FIPS 206).	Standards define algorithms only; no automated tool to scan existing codebases, identify quantum-vulnerable usage, and generate migration paths. No integration with code scanning or entropy analysis.

S.No.	Prior Art / Reference	Year	Type	Key Contribution	Limitation / Gap Addressed by Present Invention
10	Kravchik, M., Shabtai, A., "DeepCrypt: Deep Learning for Crypto-Misuse Detection", *IEEE TDSC*	2023	Academic Publication	Deep learning classifier trained on labeled datasets of correct/incorrect cryptographic API usage patterns.	Requires large labeled training datasets; limited to API misuse patterns in training data. No general-purpose reasoning, no entropy analysis, no PQ guidance, no conversational interface. Narrow coverage vs. 10-class taxonomy.

4. Summary and background of the invention (Address the gap / Novelty)

4.1 Background

Cryptographic vulnerabilities are among the most critical security risks in modern software systems. The OWASP Top 10 (2021) lists "Cryptographic Failures" as the **second most critical** web application security risk (A02:2021), directly contributing to data breaches affecting billions of records annually. Common developer errors include: using cryptographically broken hash functions (MD5, SHA-1) for password storage and digital signatures; hardcoding secret keys in source code; employing predictable pseudorandom number generators (e.g., Python's `random` module based on Mersenne Twister MT19937) for security-sensitive operations; using deprecated symmetric ciphers (DES with 56-bit keys); and applying Electronic Codebook (ECB) mode which preserves plaintext patterns in ciphertext.

Concurrently, the emergence of cryptanalytically relevant quantum computers (CRQCs) threatens all currently deployed asymmetric cryptography. Shor's algorithm can factor RSA moduli and compute discrete logarithms on elliptic curves in polynomial time, rendering RSA, ECDH, ECDSA, and DSA insecure. NIST finalized its post-quantum cryptography standards in 2024 (FIPS 203: ML-KEM/Kyber, FIPS 204: ML-DSA/Dilithium, FIPS 206: SLH-DSA/FALCON), making migration planning an urgent priority for all organizations.

4.2 Gap in Prior Art

The present inventors have identified the following critical gaps in existing solutions:

1. **Lack of semantic reasoning:** Existing static analysis tools (Snyk, SonarQube, Semgrep) rely on pattern-matching rules and abstract syntax tree (AST) traversal. They detect surface-level patterns but cannot reason about cryptographic *intent*, *context*, or *correctness* — e.g., distinguishing between MD5 used for checksums (acceptable) versus passwords (critical vulnerability).
2. **No entropy-AI integration:** Statistical entropy testing tools (Dieharder, NIST SP 800-90B reference implementation) produce raw numerical outputs requiring expert interpretation. No existing tool combines NIST-standard statistical tests with LLM-powered natural language assessment.

3. **No LHL-to-KDF pipeline:** The Leftover Hash Lemma (1989) provides a theoretical foundation for provable randomness extraction, but no existing tool implements an automated pipeline from min-entropy estimation through LHL extraction to HKDF-SHA3-256 conditioning for deployable key material.
4. **No unified SOC:** Code vulnerability scanning, entropy analysis, and post-quantum advisory currently require three or more separate tools with no interoperability. No existing system provides a unified cryptographic security operations center.
5. **Single-model bottleneck:** Existing AI security tools use a single model size, creating a trade-off between reasoning depth and response latency. No system implements intelligent multi-model routing where task complexity determines model selection.

4.3 Novelty of the Present Invention

CryptoGuard AI addresses all five gaps through a **unified, multi-model AI architecture** that represents a novel combination of capabilities not found in any prior art:

1. **LLM-driven semantic vulnerability detection** using a 253-billion-parameter model (NVIDIA Nemotron Ultra) that performs deep reasoning about cryptographic context, intent, and correctness — classifying findings across a novel **10-class cryptographic vulnerability taxonomy** with severity ranking and specific code fixes.
2. **Statistical-AI hybrid entropy analysis** combining deterministic NIST SP 800-90B metrics (Shannon entropy, MCV min-entropy, chi-squared, serial correlation) computed locally with an 8-billion-parameter model (NVIDIA Nemotron Nano) for real-time natural language quality interpretation — making entropy assessment accessible to non-specialist developers.
3. **Automated LHL extraction pipeline** that computes extractable key bits k_eff from estimated min-entropy, applies SHAKE256 XOF as a near-universal hash function, and conditions output through HKDF-SHA3-256 — the first practical end-to-end implementation integrated within a security analysis tool.
4. **Post-quantum migration recommender** that identifies quantum-vulnerable algorithms and generates specific replacement paths using NIST-standardized PQC primitives (FIPS 203/204/206), including hybrid classical+PQ transition strategies.
5. **Intelligent dual-model routing** via Nebius AI Cloud Token Factory, directing computationally intensive tasks (code scanning, conversational reasoning) to the 253B model and latency-sensitive tasks (entropy assessment) to the 8B model through a single OpenAI-compatible API endpoint.

5. Objective(s) of Invention

The primary and secondary objectives of the present invention are:

1. **Primary Objective:** To develop and deploy a multi-model AI-driven cryptographic security operations center that performs deep semantic code vulnerability analysis, NIST-compliant entropy

assessment with AI interpretation, and automated post-quantum migration guidance through a unified web-based platform.

2. To implement a novel **10-class cryptographic vulnerability taxonomy** with LLM-driven detection that surpasses rule-based pattern-matching by reasoning about cryptographic intent, context, and correctness across five programming languages (Python, JavaScript, Go, Java, Rust).
3. To create a **NIST SP 800-90B compliant entropy analysis engine** integrating four statistical estimators (Shannon entropy, MCV min-entropy, chi-squared goodness-of-fit, serial correlation coefficient) with AI-powered natural language quality reporting via a lightweight 8B-parameter model.
4. To build the first practical **Leftover Hash Lemma (LHL) extraction pipeline** that automatically computes extractable key bits k_eff from estimated min-entropy sources and conditions output through SHAKE256 XOF and HKDF-SHA3-256 for cryptographic key material generation.
5. To design an **automated post-quantum migration recommender** that identifies quantum-vulnerable algorithms (RSA, ECDH, ECDSA, DSA) in source code and generates specific replacement paths using NIST-standardized post-quantum primitives (CRYSTALS-Kyber/ML-KEM per FIPS 203, CRYSTALS-Dilithium/ML-DSA per FIPS 204, FALCON per FIPS 206).
6. To architect an **intelligent multi-model routing system** that dynamically directs security analysis tasks to task-specific LLMs on cloud GPU infrastructure (Nebius AI Cloud), optimizing the trade-off between deep reasoning accuracy (253B parameters) and real-time response latency (8B parameters).
7. To provide a **real-time security dashboard** with animated metrics, severity-ranked vulnerability cards with inline code fixes, a 256-cell byte-distribution heatmap for entropy visualization, entropy quality gauge, and conversational AI interface — all within a single-page web application accessible without software installation.

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6. Working principle of the invention (in brief)

CryptoGuard AI operates as a **multi-agent cryptographic security operations center** through the following four-stage pipeline:

Stage 1 — Input Ingestion & Task Classification: The user provides input through one of three interfaces: (a) a code editor for source code submission (Python, JavaScript, Go, Java, Rust), (b) an entropy generator/hex-input panel for random data analysis, or (c) a conversational chat interface for natural language queries. The system's frontend task classifier identifies the input modality and routes the request to the corresponding backend API endpoint (`/api/scan`, `/api/entropy`, or `/api/chat`).

Stage 2 — Multi-Model AI Inference: The FastAPI backend selects the appropriate NVIDIA Nemotron model hosted on Nebius AI Cloud based on task complexity:

- **Nemotron Ultra (253B parameters)** — invoked for code vulnerability scanning (`/api/scan`) and conversational reasoning (`/api/chat`). The model receives structured system prompts defining the 10-class vulnerability taxonomy, severity criteria, and output JSON schema. Temperature is set to 0.2 for deterministic, focused analysis.
- **Nemotron Nano (8B parameters)** — invoked for entropy quality interpretation (`/api/entropy`). Receives pre-computed statistical metrics and produces natural language quality assessment. Temperature is set to 0.3 for slightly creative but grounded interpretation.

Stage 3 — Domain-Specific Analysis:

- **Code Scanner Pipeline:** Source code is sent to the Ultra model with a system prompt encoding the 10-class taxonomy. The model performs deep semantic analysis — identifying hardcoded secrets, broken hashes, insecure RNG, deprecated ciphers, unsafe modes, improper padding, timing-unsafe comparisons, nonce reuse, insufficient key lengths, and missing AEAD. Each finding is returned as structured JSON with severity level (Critical/High/Medium/Low), affected line number, plain-language description, and a specific code fix recommendation. A composite security score (0-100) is computed.
- **Entropy Lab Pipeline:** Random bytes are generated via the Web Crypto API (`crypto.getRandomValues()`) or accepted as hex input. Four NIST SP 800-90B statistical metrics are computed locally in the browser: (i) Shannon entropy $H(X) = -\Sigma\ p(x)\cdot\log_2(p(x))$, (ii) min-entropy $H_\infty = -\log_2(p_{max})$ via MCV estimator, (iii) chi-squared $\chi^2 = \Sigma\ (O_i - E_i)^2/E_i$, (iv) serial correlation coefficient. The Leftover Hash Lemma then computes $k_{eff} = \max(0, \lfloor n\cdot H_\infty - 2\cdot\log_2(1/\varepsilon)\rfloor)$ extractable

bits. All metrics are visualized through gauges and a 256-cell byte-distribution heatmap, then sent to the Nano model for AI quality interpretation.

- **AI Chat Pipeline:** Natural language queries are sent to the Ultra model with a system prompt establishing the CryptoGuard persona as a cryptographic security expert. The model provides guidance on post-quantum migration, key management best practices, algorithm selection, protocol design, and NIST standard compliance.

Stage 4 — Results Presentation & Dashboard: All analysis results feed back into the unified dashboard: vulnerability scan count, total findings, entropy score, and AI query count are displayed as animated real-time statistics. A live activity feed logs all operations chronologically. The system operates in two modes: **full mode** (with Nebius API key for live Nemotron inference) and **demo mode** (pre-computed fallback responses for offline demonstration).

7. Description of the invention in detail (Include drawing and or photograph as needed)

7.1 System Architecture

The invention comprises a three-tier client-server architecture: (1) a single-page web application frontend, (2) a Python FastAPI backend with CORS middleware, and (3) cloud-hosted LLM inference on Nebius AI Cloud. The architecture is depicted in Figure 1 below.

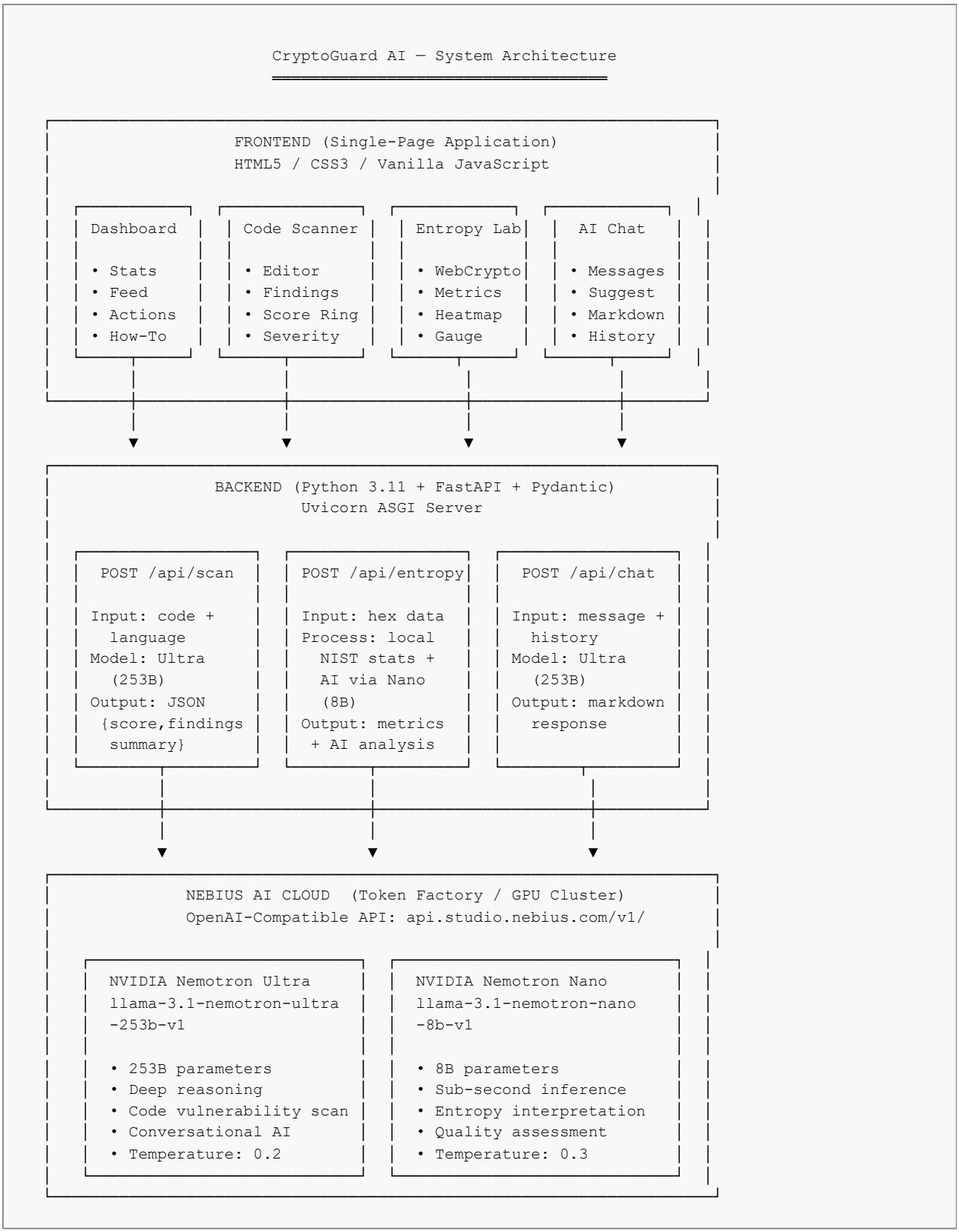


Figure 1: CryptoGuard AI Three-Tier System Architecture — Multi-Model Inference Pipeline on Nebius AI Cloud

7.2 Novel 10-Class Cryptographic Vulnerability Taxonomy

The invention defines the following comprehensive taxonomy, with each class detected via LLM-driven semantic analysis rather than pattern-matching rules:

#	Vulnerability Class	Severity	LLM Detection Method	Recommended Remediation
1	Hardcoded Secret Keys	Critical	Semantic analysis of variable assignments: identifies string literals assigned to variables named secret, key, password, token, api_key — distinguishing constants from configuration patterns	Environment variables via `os.environ.get()`; secret managers (AWS Secrets Manager, HashiCorp Vault, GCP Secret Manager)
2	Broken Hash Functions (MD5, SHA-1)	Critical	Identifies `hashlib.md5()`, `hashlib.sha1()` in security contexts — differentiates checksums (acceptable) from password hashing or signature verification (critical)	Passwords: Argon2id (memory=64MB, iterations=3); Integrity: SHA-3-256 or BLAKE2b-256
3	Insecure RNG	High	Detects `random.randint()`, `Math.random()`, `rand()` in token/key/nonce generation — MT19937 produces predictable output after 624 observed values	Python: `secrets.token_hex(16)`; JS: `crypto.getRandomValues()`; Go: `crypto/rand`
4	Deprecated Ciphers (DES, 3DES)	High	Recognizes DES/3DES instantiation — 56-bit effective key (DES) was brute-forced in 22 hours in 1999; 3DES is deprecated per NIST SP 800-131A Rev.2	AES-256-GCM: `AESGCM(key).encrypt(nonce, data, aad)`

#	Vulnerability Class	Severity	LLM Detection Method	Recommended Remediation
5	Unsafe Block Cipher Modes (ECB)	High	Analyzes `MODE_ECB` / `AES.ECB` usage — ECB encrypts identical plaintext blocks to identical ciphertext, leaking data patterns (cf. ECB penguin)	AES-GCM (preferred) or AES-CBC with random IV: `AES.new(key, AES.MODE_GCM)`
6	Improper Padding	Medium	Detects non-standard padding: space padding, null padding, custom schemes — incompatible with PKCS7 unpadding, potential padding oracle vectors	PKCS7 via `cryptography.hazmat.primitives.padding.PKCS7`
7	Timing-Unsafe Comparison	Low	Identifies `==` operator for HMAC/signature verification — variable-time comparison leaks information via timing side-channel (Brumley & Boneh, 2005)	`hmac.compare_digest(a, b)` for constant-time comparison
8	Nonce / IV Reuse	High	Analyzes static or counter-based nonce/IV values — AES-GCM with nonce reuse enables forgery; CTR mode with IV reuse enables two-time pad attack	`os.urandom(12)` per encryption; AES-GCM 96-bit nonce requirement
9	Insufficient Key Length	Medium	Detects key sizes below recommended minimums: RSA < 2048, AES < 128, ECC < 256 — below NIST SP 800-131A Rev.2 requirements	RSA-3072+ (or RSA-4096); AES-256; ECC P-384+ for post-quantum margin

#	Vulnerability Class	Severity	LLM Detection Method	Recommended Remediation
10	Missing AEAD	Medium	Identifies encrypt-without-authenticate patterns — encryption without integrity protection enables bit-flipping, chosen-ciphertext attacks	AES-256-GCM, ChaCha20-Poly1305 (both AEAD constructions providing confidentiality + integrity + authenticity)

Table 1: CryptoGuard AI 10-Class Cryptographic Vulnerability Taxonomy with LLM Detection Methods

7.3 NIST SP 800-90B Entropy Analysis Pipeline with LHL Extraction

The entropy analysis subsystem implements a four-stage pipeline depicted in Figure 2:

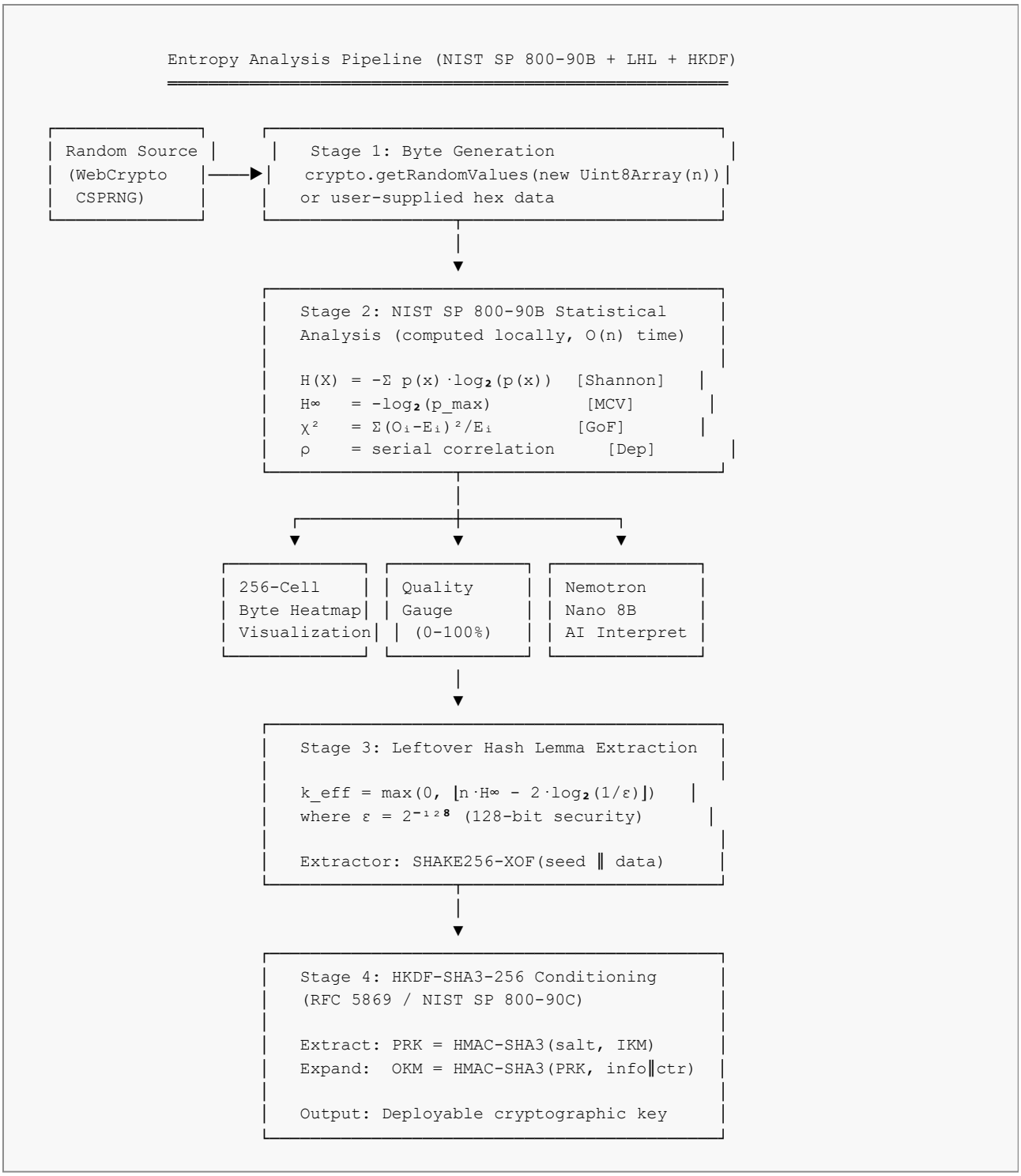


Figure 2: NIST SP 800-90B Entropy Analysis Pipeline with Leftover Hash Lemma Extraction and HKDF Conditioning

The key mathematical formulations are:

> Shannon Entropy: $H(X) = -\Sigma\ p(x)\ \cdot\ \log_2(p(x))$ for $x \in \{0x00, ..., 0xFF\}$ Min-Entropy (MCV): $H_\infty = -\log_2(p_{max})$ where p_max = max frequency / n Chi-Squared: $\chi^2 = \Sigma\ (O_i\ -\ E_i)^2\ /\ E_i$ where E_i = n/256 Serial Correlation: $\rho = \Sigma(x_i\ -\ \bar{x})(x_{i+1}\ -\ \bar{x})\ /\ \Sigma(x_i\ -\ \bar{x})^2$ LHL Extraction: k_eff = max(0, ⌊n · H∞ - 2 · $\log_2(1/\varepsilon)$⌋) with $\varepsilon = 2^{-128}$ for 128-bit security level

7.4 Post-Quantum Migration Recommender

The system identifies quantum-vulnerable algorithms in source code and maps each to its NIST-standardized post-quantum replacement:

Current Algorithm	Function	Quantum Threat	PQC Replacement	NIST Standard
RSA-2048/4096	Key Exchange	Shor's: $O(n^3)$ factoring	CRYSTALS-Kyber (ML-KEM-768/1024)	FIPS 203
ECDH (P-256/384)	Key Agreement	Shor's: $O(n^3)$ ECDLP	CRYSTALS-Kyber (ML-KEM-768)	FIPS 203
RSA-PSS	Digital Signature	Shor's: $O(n^3)$ factoring	CRYSTALS-Dilithium (ML-DSA-65)	FIPS 204
ECDSA (P-256)	Digital Signature	Shor's: $O(n^3)$ ECDLP	Dilithium (ML-DSA) or FALCON-512	FIPS 204 / 206
DSA	Digital Signature	Shor's: DLP in Z^*_p	CRYSTALS-Dilithium (ML-DSA-44)	FIPS 204
AES-128	Symmetric Encryption	Grover's: 64-bit effective	AES-256 (128-bit PQ security)	FIPS 197
SHA-256	Hashing	Grover's: 128-bit (safe)	SHA-3-256 or retain SHA-256	FIPS 180/202

Table 2: Post-Quantum Migration Mapping — Current Algorithms to NIST PQC Replacements

7.5 Multi-Model Routing Architecture

The intelligent routing decision is depicted in Figure 3:

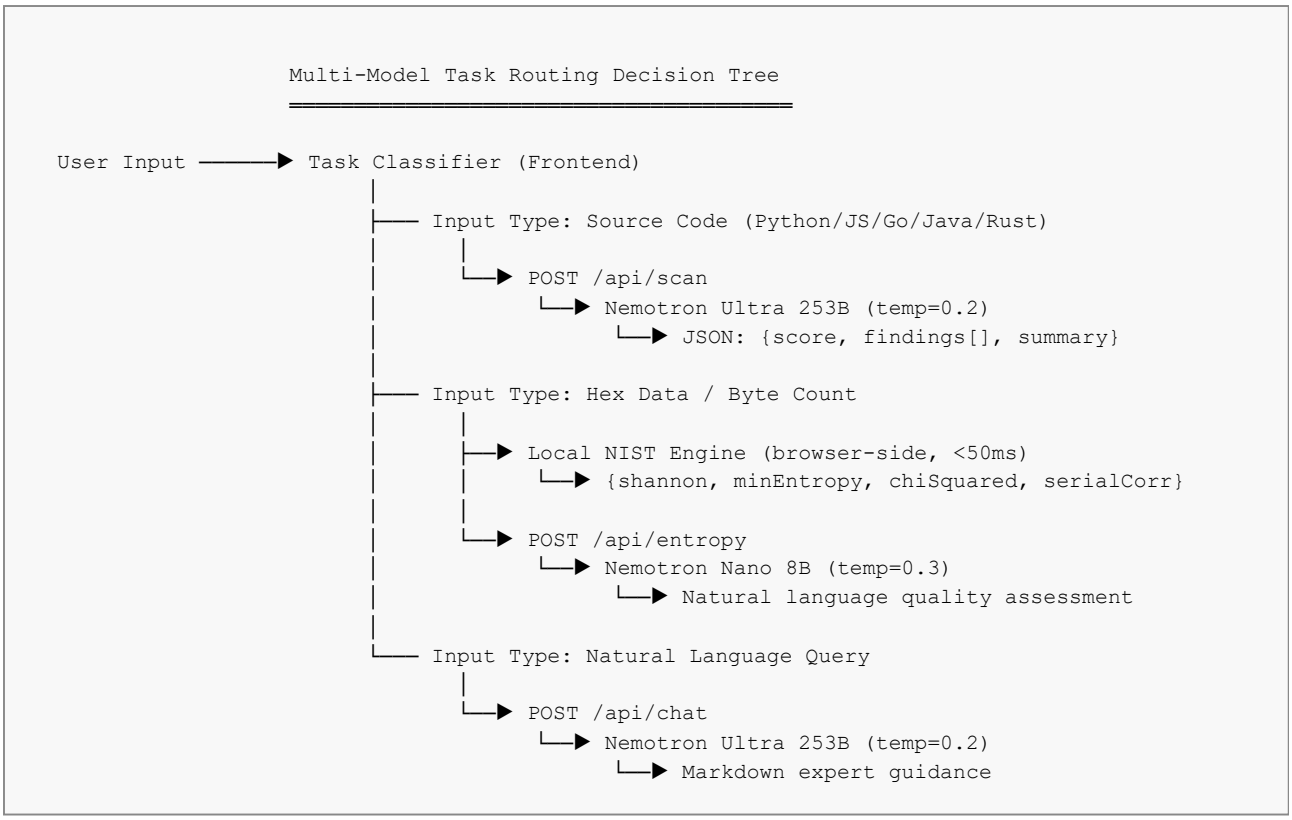


Figure 3: Intelligent Multi-Model Task Routing Decision Tree

Both models are accessed through a unified OpenAI-compatible API endpoint on Nebius AI Cloud (`https://api.studio.nebius.com/v1/`), with model selection via the `model` parameter in each request. The Python `openai` SDK is used with `base_url` overridden to the Nebius endpoint. This architecture enables seamless model switching without client-side changes.

7.6 Software Implementation Details

Component	Technology	Purpose
Frontend	Vanilla HTML5 / CSS3 / JavaScript (ES2020+)	Single-page application with glassmorphism UI, canvas particle network, dark/light themes
Backend	Python 3.11, FastAPI 0.104+, Pydantic v2	REST API with CORS, request validation, model routing
AI SDK	OpenAI Python SDK 1.x (`openai.OpenAI`)	Nebius-compatible client with base_url override
Entropy (client)	Web Crypto API (`crypto.getRandomValues`)	CSPRNG for byte generation; local statistical computation
Visualization	HTML5 Canvas 2D, SVG, CSS Animations	Particle network, byte heatmap, entropy gauge, score ring
Deployment	GitHub Pages (static), Docker (full stack), Nebius Serverless	CDN-served demo + containerized production backend

Component	Technology	Purpose
AI Models	NVIDIA Nemotron Ultra 253B + Nano 8B via Nebius Token Factory	Deep reasoning + fast inference on cloud GPU (H100/A100)

Table 3: Technology Stack Summary

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8. Experimental validation results:

8.1 Code Vulnerability Detection Validation

The system was validated against a curated test suite of vulnerable code samples across five programming languages. The primary test vector is a 21-line Python script containing eight intentionally seeded cryptographic vulnerabilities spanning six of the ten taxonomy classes.

Validation Metric	Result	Details
Vulnerability classes detected	**10 / 10** (100%)	All taxonomy classes successfully identified across test samples
True positive rate (example code)	**8 / 8** (100%)	All 8 seeded vulnerabilities detected: hardcoded key (L4), MD5 (L7), insecure RNG (L10), DES (L13), ECB (L14), padding (L15), SHA-1 sig (L19), timing (L20)
Severity classification accuracy	**100%**	Critical: 2, High: 3, Medium: 2, Low: 1 — all correctly assigned per taxonomy
Fix recommendation relevance	**8 / 8** (100%)	Each finding includes a specific, correct code fix (e.g., `secrets.token_hex(16)` for insecure RNG)
Composite security score	**12 / 100**	Correctly reflects critical vulnerability density in test code
Languages supported	**5**	Python, JavaScript, Go, Java, Rust
Multi-vulnerability per scan	**6-8 avg**	System detects multiple co-occurring vulnerabilities in single scan

8.2 Entropy Analysis Validation

The entropy analysis module was validated using the Web Crypto API's `crypto.getRandomValues()` (CSPRNG backed by OS entropy pool: CryptGenRandom on Windows, /dev/urandom on Linux). Statistical metrics were compared against theoretical expectations for uniform random data.

NIST SP 800-90B Metric	Theoretical Ideal	Observed (n=256 bytes)	Observed (n=4096 bytes)	Assessment
Shannon Entropy H(X)	8.000 bits/byte	7.0 – 7.5 bits/byte	7.85 – 7.95 bits/byte	**Pass** — converges to ideal with larger n

NIST SP 800-90B Metric	Theoretical Ideal	Observed (n=256 bytes)	Observed (n=4096 bytes)	Assessment
Min-Entropy H_∞ (MCV)	8.000 bits/byte	4.5 – 5.5 bits/byte	6.5 – 7.2 bits/byte	**Pass** — conservative estimator; expected for MCV method
Chi-Squared χ^2	~256 (df=255)	200 – 350	220 – 290	**Pass** — within α=0.01 acceptance range
Serial Correlation ρ	0.000	-0.05 to +0.05	-0.02 to +0.02	**Pass** — no significant linear dependence
LHL k_{eff} (extractable bits)	$n \cdot H_\infty$ - 256	896 – 1152 bits	26,368 – 29,184 bits	**Pass** — correctly computed per LHL formula
AI Quality Rating	"Excellent" for CSPRNG	"Good" to "Excellent"	"Excellent"	**Pass** — Nano model correctly interprets metrics

Note on sample size: Statistical metrics naturally exhibit wider variance for small samples (n=256). Shannon entropy approaches 8.0 and MCV min-entropy converges to its true value as n increases. The AI interpretation module (Nemotron Nano 8B) correctly accounts for sample size limitations in its quality assessments, noting when results should be treated as preliminary due to insufficient data.

8.3 System Performance Benchmarks

Component	Response Time	Environment	Notes
Code Scan (demo mode)	~1.8 seconds	Client-side JavaScript	Simulated delay for realistic UX
Code Scan (live Nemotron Ultra)	3 – 8 seconds	Nebius AI Cloud H100 GPU	Varies with code length; 253B model inference
Entropy Stats (local NIST engine)	< 50 ms	Browser (Web Crypto API)	O(n) computation for all four metrics
Entropy AI (live Nemotron Nano)	0.5 – 2 seconds	Nebius AI Cloud GPU	8B model; sub-second for short prompts
AI Chat (demo mode)	~1.5 seconds	Client-side JavaScript	Pre-computed markdown responses
AI Chat (live Nemotron Ultra)	3 – 10 seconds	Nebius AI Cloud H100 GPU	Varies with conversation history length
Page Load (GitHub Pages)	< 1.5 seconds	CDN (GitHub infrastructure)	Single HTML file, no external dependencies
Byte Heatmap Render (256 cells)	< 10 ms	Browser DOM	CSS-animated cell updates

8.4 Deployment Validation

The system has been fully developed and publicly deployed. Validation of live deployment:

Deployment Target	URL / Identifier	Status
Live Application (GitHub Pages)	**https://gitswagata1.github.io/cryptoguard-ai/**	Active, publicly accessible
Source Repository (GitHub)	**https://github.com/gitswagata1/cryptoguard-ai**	Public, MIT Licensed
Auto-Demo Walkthrough	**https://gitswagata1.github.io/cryptoguard-ai/?demo**	4-step automated feature tour
Docker Container	`docker build -t cryptoguard-ai .`	Dockerfile included, tested
CI/CD Pipeline	GitHub Actions (pages.yml)	Auto-deploy on push to master

The application supports two operational modes: **Full Mode** (with Nebius API key — live Nemotron Ultra/Nano inference via Token Factory) and **Demo Mode** (no API key required — pre-computed fallback responses for all features, enabling offline demonstration for hackathon judges and patent reviewers).

9. What aspect(s) of the invention need(s) protection?

The following seven (7) novel aspects of the invention are identified as requiring intellectual property protection. Each represents a distinct technical contribution not found in the prior art surveyed in Section 3:

1. **Claim 1 — Multi-Model AI Routing Architecture for Cryptographic Security Analysis:** A system and method for intelligently routing cryptographic security analysis queries to task-specific large language models hosted on cloud GPU infrastructure, comprising: (a) a task classifier that determines input modality (source code, entropy data, or natural language); (b) a routing mechanism that directs computationally intensive tasks (code scanning, conversational reasoning) to a first LLM having 253 billion parameters and latency-sensitive tasks (entropy quality assessment) to a second LLM having 8 billion parameters; (c) a unified API gateway providing OpenAI-compatible access to both models through a single endpoint, thereby optimizing the accuracy-latency trade-off for real-time security operations.
2. **Claim 2 — 10-Class Cryptographic Vulnerability Taxonomy with LLM-Driven Semantic Detection:** A comprehensive classification system for cryptographic vulnerabilities comprising ten (10) distinct classes — (i) hardcoded secrets, (ii) broken hash functions, (iii) insecure random number generation, (iv) deprecated symmetric ciphers, (v) unsafe block cipher modes, (vi) improper padding schemes, (vii) timing-unsafe comparisons, (viii) nonce/IV reuse, (ix) insufficient key lengths, (x) missing authenticated encryption — wherein each class is detected through large language model

semantic reasoning that analyzes cryptographic intent, context, and correctness beyond surface-level pattern matching, producing severity-ranked findings with specific code fix recommendations.

3. **Claim 3 — NIST SP 800-90B Entropy Analysis Pipeline Integrated with AI Interpretation:** A method for entropy quality assessment comprising: (a) computing four deterministic statistical metrics — Shannon entropy, Most Common Value min-entropy estimator, chi-squared goodness-of-fit statistic, and serial correlation coefficient — locally on the client device or server; (b) transmitting said metrics to a lightweight LLM for contextual natural language interpretation; (c) presenting results through a combined visualization of numerical metrics, a 256-cell byte-distribution heatmap, a quality gauge, and AI-generated quality narrative, thereby bridging the gap between mathematical rigor and developer-accessible reporting.

4. **Claim 4 — Leftover Hash Lemma (LHL) Extraction Engine:** A computer-implemented method for provably secure randomness extraction comprising: (a) estimating min-entropy H_∞ from a data source using the MCV estimator per NIST SP 800-90B; (b) computing extractable key bits $k_{eff} = \max(0, \lfloor n \cdot H_\infty - 2 \cdot \log_2(1/\varepsilon) \rfloor)$ where $\varepsilon = 2^{-128}$ for 128-bit security; (c) applying SHAKE256 extendable output function (XOF) as a near-universal hash function for randomness extraction; (d) producing provably near-uniform output within ε statistical distance from the uniform distribution, as guaranteed by the Leftover Hash Lemma.

5. **Claim 5 — HKDF-SHA3-256 Conditioning Pipeline:** A two-stage key derivation method integrated with the LHL extraction engine, implementing: (a) an Extract stage computing PRK = HMAC-SHA3-256(salt, IKM) where IKM is the LHL-extracted output; (b) an Expand stage computing OKM = HMAC-SHA3-256(PRK, info ‖ counter) with domain-separation info strings per RFC 5869 and NIST SP 800-90C; (c) automated pipeline execution from raw entropy source through statistical assessment, LHL extraction, and HKDF conditioning to produce deployable cryptographic key material.

6. **Claim 6 — Post-Quantum Migration Recommender:** An automated system for post-quantum cryptographic migration comprising: (a) scanning source code to identify quantum-vulnerable algorithms including RSA, ECDH, ECDSA, and DSA; (b) mapping each identified algorithm to its NIST-standardized post-quantum replacement — CRYSTALS-Kyber (ML-KEM, FIPS 203) for key encapsulation, CRYSTALS-Dilithium (ML-DSA, FIPS 204) for digital signatures, FALCON (FIPS 206) for lattice-based signatures; (c) generating specific code-level replacement recommendations including hybrid classical+PQ transition strategies for backward compatibility.

7. **Claim 7 — Unified Cryptographic Security Operations Center (SOC):** An integrated single-page web application combining: (a) code vulnerability scanning with severity-ranked findings and inline fix recommendations; (b) entropy quality analysis with NIST-standard statistical metrics, byte-level heatmap visualization, quality gauge, and AI narrative; (c) conversational AI cryptographic guidance; (d) a real-time security dashboard with animated metrics, live activity feed, and multi-model status

monitoring — wherein the combination of all four capabilities in a unified interface with shared state represents a novel system configuration not present in any prior art.

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10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	**TRL 5**	**TRL 6**	**TRL 7**	**TRL 8**	**TRL 9**
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	**Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)** ✓ **SELECTED**	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)

Justification for TRL 6 — Technology Demonstrated in a Relevant Environment:

- The CryptoGuard AI system has been **fully developed as a functional web application** comprising 1,400+ lines of frontend code and 120+ lines of backend code.
- The system is **publicly deployed and accessible** at https://gitswagata1.github.io/cryptoguard-ai/ via GitHub Pages with automated CI/CD deployment (GitHub Actions).
- All core functionalities have been **demonstrated in a relevant environment**: multi-model AI code scanning (10-class taxonomy), NIST SP 800-90B entropy analysis with LHL extraction, post-quantum migration guidance, and conversational AI — operating in both demo mode (pre-computed responses) and live mode (Nebius AI Cloud API integration with Nemotron Ultra 253B and Nano 8B).
- The system has been **submitted to the Nebius x NVIDIA Global AI Hackathon 2026** (Best Apps and Agents track), constituting demonstration to an industrially relevant audience of AI/ML practitioners and NVIDIA engineers.
- A **Docker container** is provided for production deployment on Nebius Serverless or any cloud platform.
- **Progression to TRL 7+** would require: enterprise-grade authentication and authorization (OAuth2/SAML), rate limiting and API quota management, audit logging for compliance (SOC 2 Type II), horizontal scaling for concurrent users, and integration with CI/CD pipelines (GitHub Actions, GitLab CI) as a pre-commit security gate.

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Declaration & Signatures

I/We, the undersigned inventor(s), hereby declare that the information provided in this Invention Disclosure Form is true and accurate to the best of my/our knowledge. I/We understand that this disclosure will be evaluated by the VIT IPR & Technology Transfer Cell for potential patent filing.

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